S7 Formal Verification of Selected NDT Mathematical Components

Machine-checked boundaries from reduced state to licensed finite-horizon geometry

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Abstract This note records a bounded Lean formalization of selected mathematical components used by Noetic Diffusion Theory (NDT). The formal development starts with a finite counterexample for reduced-state insufficiency and continues through covariance propagation, finite-horizon nondegeneracy, positive-semidefinite factorization, and proof-carrying inverse ellipsoids. The theorems are exact under their stated finite-dimensional assumptions. They support the architecture and claim discipline of NDT; they do not validate empirical accessibility, identify a biological mechanism, or establish that an MNPS chart is a latent neural state.

1. Scope and epistemic status

This supplement is a formal verification record for selected mathematical objects in the NDT v2 article. It is an infrastructure note, not a new empirical results section and not a claim that the whole theory has been formalized. The source is the Lean project at the repository-relative path `formalization/lean`. Its Lean namespace is `NoeticDiffusion`.

The formal results answer questions of the following form: given explicit finite-dimensional assumptions, does a stated algebraic identity or logical equivalence follow? They do not answer whether the assumptions hold for neural recordings. A kernel-checked theorem can certify a conditional mathematical claim while leaving model selection, measurement validity, residual quality, and biological interpretation open.

Reading rule

Proved means that the named proposition is accepted by Lean's kernel under the hypotheses in its declaration. **Not proved** means that the current project deliberately does not claim the statement. Neither label is an empirical pass or a biological explanation.

2. The formal architecture

The development follows the same separation used in the main article: a measurement chart is an observation interface, while accessibility concerns a future transition law. The mathematical chain is:

Stage	Object	Formal content and claim ceiling
1	Reduced chart	A chart $\pi : X \to Z$ can identify present observations while losing future-relevant information. A finite Boolean counterexample and a general fibre criterion are proved.
2	Affine stochastic step	For a concrete finite family of real random variables, $AX + b + \varepsilon$ has covariance $A\ \mathrm{Cov}(X)A^\top + Q$ under explicit integrability, innovation covariance, and zero cross-covariance hypotheses.
3	Finite-horizon covariance	$C(0) = 0$ and $C(h+1) = AC(h)A^\top + Q$ have a finite Gramian form. The time-varying extension allows $A(h)$ and $Q(h)$ to change with the step.
4	Innovation factors	For explicit factors $B(h)$, the propagated factors are $\Phi(T, h+1)B(h)$. For fixed transitions and $Q = BB^\top$, the factors are $A^k B$.
5	Nondegeneracy gate	Positive definiteness is equivalent to directional detection, trivial common transpose kernel, full block-column span, and full row rank, with the relevant finite index assumptions.
6	Licensed geometry	An inverse quadratic form and inverse ellipsoid require a `PosDef` witness. Affine push-forward additionally requires an explicit `IsUnit F`; failed nondegeneracy gives no ordinary inverse-ellipsoid license.

Table 1: The formal chain from reduced state to licensed finite-horizon geometry. Each arrow carries assumptions; no arrow licenses an empirical or biological interpretation.

The chain has two distinct uses in NDT. The reduced-state results formalize a possibility claim: a present chart need not be sufficient for a future law. The covariance results formalize an algebraic construction: a finite-horizon predictive-spread object can be defined and manipulated when its transition, noise, and nondegeneracy assumptions are explicit. The latter object remains a covariance-based approximation to future spread, rather than classical control-theoretic reachability or a probability coverage statement.

3. Theorem inventory and status

The table reports the current status at the level of mathematical statements. The file names and theorem names are included so that a reader can inspect the kernel-checked source directly.

NDT-related statement	Status	Lean evidence or explicit boundary
A reduced chart can lose future-relevant information.	Proved	`ReducedState.lean`: `same_present_different_observed_future` and `no_deterministic_chart_factorization`; finite counterexample.
Exact criterion for deterministic factorization through a surjective chart.	Proved	`ReducedSufficiency.lean`: `exists_factorization_iff_of_surjective`; constancy on chart fibres is equivalent to factorization.
An affine stochastic update induces the covariance recursion.	Proved under hypotheses	`StochasticCovarianceConcrete.lean`: `covMatrix_affineNoise`; the abstract interface exposes covariance additivity and linearity as assumptions.
Finite-horizon covariance has a Gramian representation.	Proved	`Covariance.lean`, `StochasticCovarianceHorizon.lean`, and `StochasticCovarianceTimeVaryingGramian.lean`; finite algebra only, with no stability assumption.
A time-varying factor chain can be characterized as nondegenerate through collectively detecting propagated factors.	Proved	`StochasticCovarianceTimeVaryingNondegeneracy.lean` for explicit factors; `StochasticCovarianceTimeVaryingFactor.lean` extends the detection, kernel, span, and rank equivalences to arbitrary PSD $Q(h)$ by existential factor selection, with the singular boundary `covarianceTimeVarying_det_eq_zero_of_not_posDef`.
Positive definiteness, detection, common-kernel triviality, span, and full rank are equivalent in the finite factor construction.	Proved	`CovarianceNondegeneracyRectangular.lean`, `CovarianceKernelEquivalence.lean`, `CovarianceBlockSpan.lean`, and `CovarianceBlockRank.lean`.
Every finite real positive-semidefinite square matrix admits a factor $Q = BB^\top$.	Proved	`CovariancePSD.lean`: `posSemidef_exists_factor_mul_transpose`; this selects an existential factor, not a numerical algorithm.
Positive definiteness licenses inverse and determinant geometry.	Proved	`CovarianceLicensing.lean`: `posDef_isUnit_licence`, `posDef_det_pos_licence`, and `logDetOfPosDef`; the license carries its witness.
Affine, scaled, and fixed-transition inverse-ellipsoid identities are exact.	Proved under hypotheses	`CovarianceEllipsoidScaling.lean`, `CovarianceEllipsoidPushForward.lean`, and `CovarianceEllipsoidFixedTransitionCommon.lean`; scaling is positive, and affine maps carry `IsUnit F`.
An arbitrary time-varying PSD sequence is automatically assigned the factors needed by the nondegeneracy chain.	Proved	`StochasticCovarianceTimeVaryingFactor.lean`: `exists_timeVarying_factor_sequence` selects factors existentially for every PSD $Q(h)$ (no numerical algorithm); `CovarianceEllipsoidTimeVarying.lean` packages the covariance-agnostic `CovarianceGate` and its affine ellipsoid.
Chart-fibre sufficiency extends to PMF- and measure-valued future laws; a chart-matched pair can carry different stochastic laws.	Proved	`KernelSufficiency.lean`: `kernel_exists_factorization_iff_of_surjective`, `kernel_measure_exists_factorization_iff_of_surjective`, `stochastic_no_kernel_chart_factorization`; pointwise law equality, no kernel-measurability or path-space claim.
Protocol or one-step his-	Proved	`KernelSufficiency.lean`: `unaugmented_chart_insufficient`,
[illegible]	Proved	[illegible]

[illegible] proper conditional [illegible]

The rows labelled **proved under hypotheses** or **proved in part** are intentionally qualified. The formal result is stronger than an informal analogy because the hypotheses are visible in the declaration, but it does not remove the need to establish those hypotheses for a scientific application. Two further modules, `SlowBranchModel.lean` and `ClosureEpochOperator.lean`, formalize simulator-internal toy-model invariants and are deliberately omitted from this inventory.

4. Assumptions behind the proofs

The core covariance and ellipsoid modules work with finite index types and finite real matrices. The horizon is a natural number, so every Gramian is a finite sum. The fixed-transition chain uses a square state matrix A and an explicit innovation factor B; the rectangular modules allow the state and innovation dimensions to differ. These choices make the statements precise without claiming that a neural state is finite, linear, or fully observed.

The concrete affine covariance theorem carries the probabilistic assumptions that an algebraic recurrence would otherwise hide: a probability measure, coordinatewise second-moment integrability, a specified covariance for the innovation, and zero state–innovation cross-covariance. The separate abstract interface is useful for composition, but its covariance laws are structure fields. It should therefore be read as an adapter boundary rather than as an independent derivation from probability theory.

Positive semidefiniteness is enough for the covariance recursion and Gramian identities. It is not enough for inverse geometry. An inverse ellipsoid is constructed only after a `Matrix.PosDef` witness is supplied or derived from a finite block-span or full-rank gate. The affine image also requires `IsUnit F`. This proof-carrying design prevents a singular matrix from silently entering an inverse or log-determinant expression.

No long-time stability condition is needed for the finite-horizon identities. Conversely, the absence of a stability assumption does not establish a stationary law, asymptotic accessibility, or a valid infinite-horizon limit.

5. Reproducibility and axiom audit

The project pins Lean 4.24.0 and its Mathlib dependency closure through `lean-toolchain`, `lakefile.toml`, and `lake-manifest.json`. From the repository root, the reproducibility gate on Windows is:

```
cd formalization/lean
lake exe cache get Mathlib.LinearAlgebra.Matrix.PosDef
lake exe cache get Mathlib.Analysis.Matrix.Order
.\check.ps1 -LakeExe lake
```

The recorded run on 2026-09-09 completed the full build with 2,698 jobs and reported 81 public theorem audits; the run on 2026-09-12, after the round-2 gates, completed 2,750 jobs and reported 320 audits. Warnings are treated as errors. The audit rejects proof placeholders and project-defined axioms; the accepted reports use only Lean's standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. The audit is evidence about the source and environment at that revision. It is not evidence that the scientific assumptions have been measured.

The formal source is deliberately separated from the measurement pipeline. It does not change the canonical MNPS $[m, d, e]$ or 9D definitions, missingness, validity gates, provenance, estimator selection, or empirical outputs. A future integration must preserve the distinction between a computed object and a scientifically qualified object.

6. Explicit boundaries

The following conclusions remain outside the formal claim:

- An inverse covariance ellipsoid is not a probability contour unless a distribution and a coverage convention are separately supplied.
- A covariance Gramian is not structural controllability, a classical control reachable set, or a demonstrated set of biologically reachable neural states.
- A theorem about a finite real chart does not identify that chart with a latent neural state, nor does it establish a Markov property for MNPS.
- A formal covariance recurrence does not validate residual quality, temporal sampling, condition separation, neighbourhood support, or out-of-sample prediction in a recording.
- Nondegeneracy of a model covariance does not prove that an empirical accessibility contrast is non-redundant, reproducible, or mechanistically informative.
- The formalization does not establish consciousness, experience, or any unique biological implementation of NDT control.

These limits are part of the result. In the terminology of the main article, the development supports the mathematical architecture while leaving the measurement and biological gates open.

7. Toward a standalone formalization paper

The present work is best placed as a substantial supplement because its immediate contribution is to make the NDT architecture auditable. Much of the covariance algebra is classical. Its value here is that the assumptions, nondegeneracy conditions, and inverse-geometry licenses are explicit and machine checked alongside the theory's distinction between measured state and future law.

A later standalone paper would need a broader mathematical question. A useful direction is: **which dynamical descriptors can be proved equivalent, and which non-equivalences are essential, in a formal theory of accessibility?** Three theorem families provide a natural route:

1. **Reduced observation and dynamical sufficiency:** the fibre criterion now covers PMF- and measure-valued kernels and finite protocol- and history-augmented states (`KernelSufficiency.lean`); the open step is a measurable-kernel and path-space version.
2. **Finite-horizon accessibility geometry:** the time-varying factor bridge for arbitrary PSD innovation sequences is closed (`StochasticCovarianceTimeVaryingFactor.lean`); the open step is to connect the licensed covariance object to declared estimator assumptions.
3. **Formal non-equivalences:** machine-checked counterexamples now separate occupancy from reachability, support from mass, covariance spread from committor, local linearization from finite-amplitude behaviour, and asymptotic stability from transient amplification (`FuturesNonEquivalence.lean`, `JacobianReactivity.lean`, `JacobianConversion.lean`); the open step is a general committor uniqueness theorem under finite absorption and the concrete chart/mechanism instantiation of the four inferential claim classes.

The NDT article can therefore cite this supplement for the bounded formal infrastructure while a future paper develops the general theory of formally certified dynamical descriptors. That separation preserves the immediate credibility benefit without presenting mathematical verification as validation of the biology.

Suggested main-text sentence

Selected mathematical components of NDT have been formalized and kernel-checked in Lean 4 under explicit finite-dimensional and nondegeneracy assumptions; the formalization supports the theory's architecture but does not validate empirical accessibility or biological mechanism (Supplementary Note S7).

8. Data and source availability

The formal source, pinned build metadata, complete theorem inventory, and audit script are in the project at `formalization/lean`. Their public availability has not yet been verified; an accessible, revision-pinned source release is needed for independent reproduction. The source-level diary records the L11 gate and its validation evidence at `project/diary/300_20260909_lean_timevarying_nondegeneracy.md`. The present note does not introduce a dataset, fitted parameter, or empirical result.